INCH - POUND
MIL-DTL-3738M
20 November 2019
SUPERSEDING
MIL-DTL-3738L
14 April 2005

DETAIL SPECIFICATION

CLOTH, ELASTIQUE, WOOL

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for four (4) types of wool elastique cloth, for use in clothing.

1.2 Classification. This specification covers the following types and classes, as specified.

1.2.1 Types and classes.

Type I - 19 ounce (12.2 ounces per square yard)
Type II - 18 ounce (11.6 ounces per square yard)
Class 1 - 64's grade wool
Class 2 - 70's grade wool
Type III - 16 ounce (10.3 ounces per square yard)
Type IV - 19 ounce (12.2 ounces per square yard), Cadet Gray 172

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Attn: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil/. .

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-43665 - Cloth, Wool: Mothproofing Treatment of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-655 - Provisions for Evaluating Quality of Cloth, Wool, Worsted and Wool Blends

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

Rules and Regulations under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1, Gray Scale for Color Change
AATCC Evaluation Procedure 2, Gray Scale for Staining
AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale
AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles
AATCC Test Method 8 - Colorfastness to Crocking: Crockmeter Method
AATCC Test Method 15 - Colorfastness to Perspiration
AATCC Test Method 16.3 - Colorfastness to Light: Xenon Arc
AATCC Test Method 20A - Fiber Analysis: Quantitative
AATCC Test Method 81 - pH of the Water-Extract from Wet Processed Textiles
AATCC Test Method 132 - Colorfastness to Drycleaning
AATCC Test Method 158 - Dimensional Changes on Drycleaning in Perchloroethylene: Machine Method

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(Copies of these documents are available on line at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D2130	- Standard Test Method for Diameter of Wool and Other Animal Fibers by Microprojection
ASTM D3775	- Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/3776M	- Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM D5034	- Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at https://www.astm.org.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.crcpress.com.)

SDL ATLAS

Part Number 402985 – Slub/Knot Replica Set

(Replica Set is available for purchase from SDL Atlas Customer Service 1-803-329-2110)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Standard sample. Unless otherwise specified, the finished cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

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3.3 Recycled, recovered, environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Wool stock. The wool shall be fleece wool, pulled wool, or any combination thereof not lower in grade than the U.S. Standard grades specified below. The use of wool top made from laps, noils or any other wool manufacturing by-products shall be prohibited.

Type I - 70's U.S. Standard

Type II -
Class 1 - 64's U.S. Standard
Class 2 - 70's U.S. Standard

Type III - 70's U.S. Standard

Type IV - 70's U.S. Standard

3.4.1.1 Wool Content. The finished cloth shall contain not less than 95 percent wool based on the dry weight of the specimen when tested as specified in 4.6.

3.4.2 Yarn. The wool shall be spun from combed top on either the Bradford, French or American System. The yarn for both warp and filling shall be 2-ply. (NOTE: Selvage yarn: to prevent edges of all types of cloth from building up when rolled, two (2) ply yarns used in the selvage may be made of a finer count than those used in the body of the cloth).

3.5 Color. The color of the finished cloth for Types I, II and III shall be as specified in the contract (see 6.2. and 6.4). The color for Type IV cloth shall be Cadet Gray 172. The color shall be obtained by stock or top dyeing. Piece dyeing will not be permitted.

3.5.1 Visual shade matching. The color and appearance of the finished cloth shall match the standard sample when tested as specified in 4.6.

3.5.2 Colorfastness. The finished cloth for all types and classes shall conform to the colorfastness requirements listed below in Table I when tested in accordance with the methods specified in 4.6.

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TABLE I. Colorfastness requirements.

	Light (after 40 AFU or 170 kJ /(m^2nm) @ 420 nm) (min.) 1/	Perspiration (acid & alkaline) (color change and staining) (min.)	Drycleaning (color change and staining) (min.)	Crocking (wet and dry) (min.)
All colors	3-4	3-4	3-4	3.5

1/ AFU: AATCC Fading Units

3.6 Physical requirements. The finished cloth shall conform to the physical requirements listed in Table II, when tested in accordance with the test methods specified in 4.6.

TABLE II. Physical requirements.

Characteristic	Class I	Class II	Class III	Class IV
Weight, ounces (min.)				
Per linear yard (based on 56-inch cloth width)	19.0	18.0	16.0	19.0
Per square yard	12.2	11.6	10.3	12.2
Yarns per inch (min.)				
Warp	128	124	138	128
Filling	98	80	96	98
Breaking strength, pounds (min.)				
Warp	160	140	130	160
Filling	90	80	70	90
Dimensional stability, percent (max.) (after one (1) and five (5) cycles)				
Warp	5.5	6.5	6.5	5.5
Filling	3.0	4.0	3.0	3.0

3.7 Weave. The weave for the Type II shall be a steep twill as shown in Figure 1. The weave for Type I, Type II and Type IV, shall be a steep twill weave as shown in Figure 2.

3.8 Width. For Government procurements only, the width of the finished cloth shall be a minimum of 60-inches exclusive of selvage or tuck-in when woven on shuttleless looms or as specified (see 6.2).

3.9 Finish. The face shall be clear and closely sheared. The wool stock or top shall be mothproofed in the dyeing operation in accordance with MIL-C-43665.

3.10 pH. The pH of the water extract of the finished cloth shall be not less than 5.0 and not greater than 8.5 when tested as specified in 4.6.

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3.11 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.8. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.12 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put-up full width on a roll as specified in 5.1 (see 6.2).

3.13 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.14 Marking. The back of the cloth shall be marked "BACK" at both ends of each roll. The marking shall be clearly legible in any indelible marking used commercially.

3.15 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects as specified in 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurements (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.3 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color and finished defects (see 4.5), and tested for the characteristics as specified in 4.6.

4.4 Conformance inspection. Conformance inspection shall include the visual examination of 4.5 and the tests of 4.6 and 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.5 Yard-by-yard examination. Each roll in the sample shall be examined yard-by-yard (full width excluding selvage) on the face side only with both overhead and back lighting. The defects are as defined in MIL-STD-655 and only defects that are clearly noticeable at normal

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inspection distance (3-feet) shall be scored. The defects are assigned demerit points in accordance with MIL-STD-655, with the exceptions listed below:

a. The defects found shall be counted regardless of their proximity to each other, except where two (2) or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted.
b. Only coarse yarns that exceed twice the normal diameter shall be scored.
c. Mixed filling shall be scored only if attributed to shade bar resulting from wrong ply, wrong twist in the yarn or off shade yarn.
d. The defects specified in Table III shall be scored four (4) points for each yard in which they occur.

TABLE III. Defects that score four (4) points.

Defect Description
Any hole, cut, tear or abrasion mark.
Edge ravels when pulled outward.
Uneven weaving throughout.
Broken or missing yarn.
Any spot, stain or streak more than 1 inch in combined directions.
Knots greater than Fabric Defect Replica Scale Level C (see 2.3).
Slubs greater than Fabric Defect Replica Scale Level D (see 2.3).
Area of poor dye penetration, streak or shade bar.
Width not as specified within a tolerance of no less than 1/2 inch.
Objectionable odor.
Slack or tight selvages.
Any embedded foreign matter or uncleanness.
Edges rolled, folded, scalloped or corded, baggy, ridgy or wavy cloth.

4.5.1 Roll identification and marking examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-655.

4.5.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-655.

4.5.3 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-655.

4.6 End item testing. The cloth shall be tested for the characteristics listed in Table IV. The methods of testing as specified wherever applicable and as listed in Table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

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Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. End item tests.

Characteristic	Requirement Reference	Test Method
Wool Grade	3.4.1	ASTM D2130 1/
Wool content	3.4.1.1	AATCC 20A (see 6.6)
Yarn ply	3.4.2	Visual 2/
Weave	3.7	Visual 2/
Mothproofing	3.9	3/
Visual shade matching	3.5.1	4.7.1
Colorfastness to: Perspiration Light Crocking Drycleaning	Table I	 AATCC 15 4/ AATCC 16.3, Opt 3 5/ AATCC 8, 6/ AATCC 132, 4/
Weight	Table II	ASTM D3776/D3776M (Method C)
Yarns per inch	Table II	ASTM D3775
Breaking strength	Table II	ASTM D5034
Dimensional stability (after one (1) and five (5) cycles)	Table II	AATCC 158
pH	3.10	AATCC 81
Toxicity	3.11	4.8

1/ Testing will be conducted on a scoured form. If any specimen fails to meet the specified grade. The lot shall be rejected.
2/ One determination per sample unit and the results are reported as “pass’ or “fail”.
3/ As specified in MIL-C-43665.
4/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.
5/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change
6/ Rated using the AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale

4.7 Methods of inspection.

4.7.1 Visual shade matching. The color and appearance of the cloth shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option C (see 6.5), with sources simulating artificial daylight D75 illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856 (± 200) K.

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4.8 Toxicity test. When required, (see 6.2) an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If toxicity requirement (see 3.11) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use garrison caps and dress uniforms.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2).
d. When first article is required (see 3.1, 4.3, 6.3)
e. Color required (see 3.5)
f. Conformance inspection acceptance quality limits (AQLs)(see 4.4)
g. When toxicity testing is required (see 3.11, 4.8)
h. Width of cloth required (see 3.8)
i. Length required if other than specified (see 3.12)
j. Inspection conditions (see 4.2)
k. Packaging (see 5.1)

6.3 First article. When a first article inspection is required (see 6.2), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.4 Standard sample. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Visual shade matching. In 2019, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be that the specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens.

6.6 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.7 Replica set. The SDL Atlas Slub/Knot Replica Set is not available from their website at this time. Recommend calling customer service, 1-803-329-2110.

6.8 Changes from previous issue. Marginal changes are not used in this revision to identify changes with respect to the previous issue due to extent of the changes.

6.9 Subject term (key word) listing.

Steep twill
Grade

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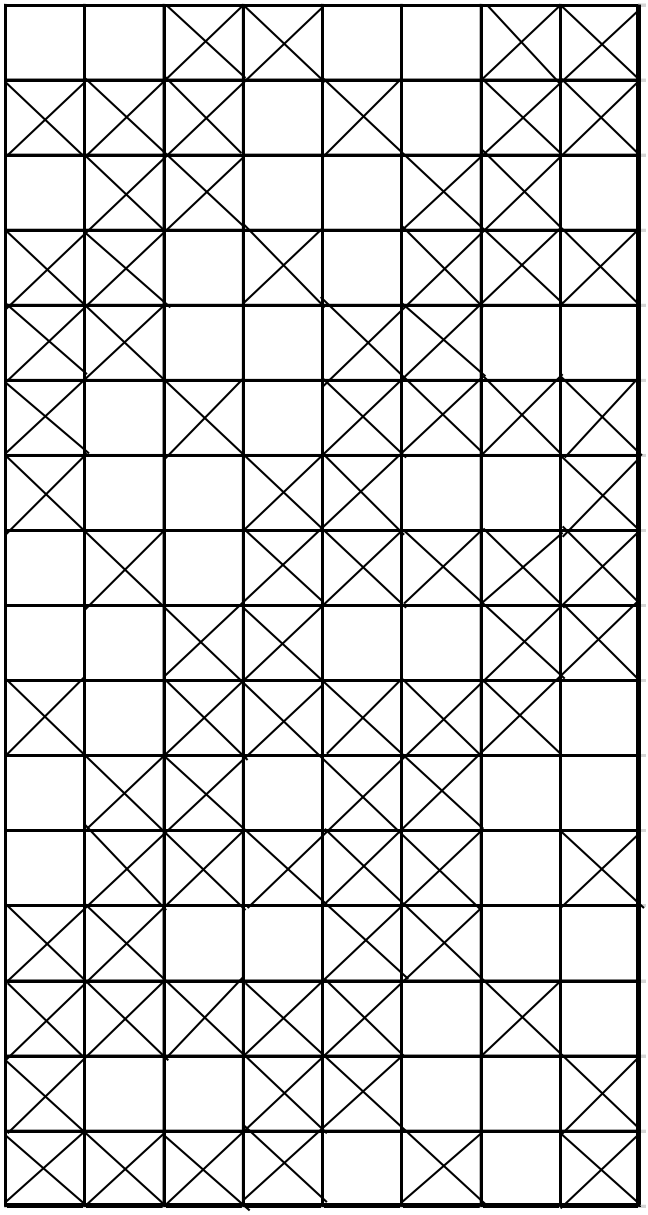


FIGURE 1. Cloth, Wool Elastique, Type II.

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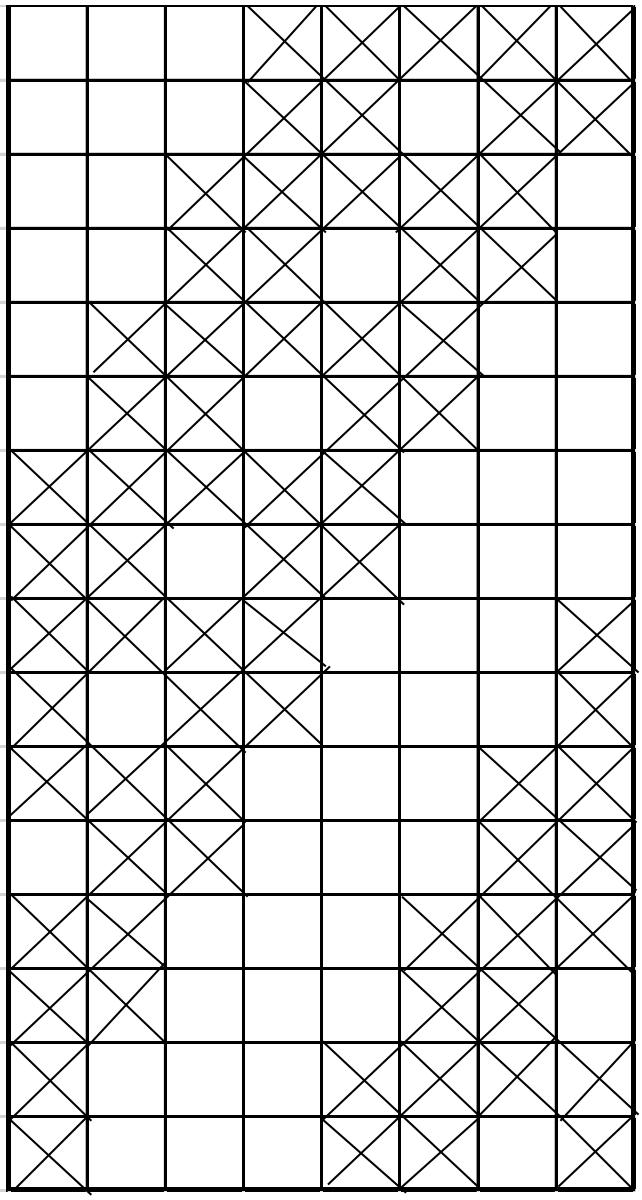


FIGURE 2. Cloth, Wool, Elastique, Type I, Type III and Type IV.

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